

Practical 12 - Development and evaluation of risk prediction models

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

SOLUTIONS

Datasets required: `rotterdam_p10.dta`

Introduction

In this practical we will perform analyses similar to those shown in the lecture, using the Rotterdam breast cancer data. The aim is to develop a prediction model for 5-year risk of the composite outcome of disease recurrence or death from any cause, and to evaluate the predictive performance of the model.

The data used is the same as in Practicals 10 and 11. The variables we will use are summarised in the table below.

Variable name	Description
<code>pid</code>	Patient identifier
<code>year_cat</code>	Year of surgery. 0: before 1990, 1: 1990 or later
<code>age</code>	Age at surgery
<code>meno</code>	Menopausal status. 0=premenopausal, 1=postmenopausal
<code>size</code>	Tumour size: categorical with levels ≤ 20 mm, 20 – 50mm, > 50 mm
<code>grade</code>	Differentiation grade: categorical with levels 2 and 3
<code>enodes</code>	Number of positive lymph nodes, transformed as $\text{enodes} = \exp(-0.12 * \text{nodes})$

Variable name	Description
<code>pgr</code>	Progesterone receptors (fmol/l)
<code>er</code>	Estrogen receptors (fmol/l)
<code>chemo</code>	Chemotherapy: 0=no, 1=yes
<code>hormon</code>	Hormonal treatment: 0=no, 1=yes
<code>time</code>	Years to the first of recurrence or death without recurrence
<code>outcome</code>	Indicator of composite outcome: 0=censored, 1=event
<code>outcome_cr</code>	Competing events outcome: 0=censored, 1=recurrence, 2=death without recurrence

Aims

The aims of this session are:

- Fit prediction models and evaluate their performance.
- Evaluate prediction model performance using measures of calibration, discrimination, and overall performance.
- Use random split-sampling and k-fold cross-validation for evaluation of prediction model performance.

Question 1

- Load the data.
- It turns out to be sensible to change the scale of the variables `pgr` and `er`, so we will divide each of these by 100 for this analysis.
- Create a training set on which the model will be fitted, and a test set on which the predictive performance of the model will be evaluated. The training set is a random sample of 70% of the cohort, and the remaining 30% for the test set.

Question 2

- Use `stset`.
- Using the training set, fit a Cox regression for the composite outcome using the following predictors: `hormon`, `chemo`, `age`, `meno`, `size`, `grade`, `enodes`, `pgr100`, `er100`, `year_cat`.
- Use the model to obtain estimates of 5-year risk for individuals in the test set.
- What is the range of the predicted risks in the test set?

SOLUTIONS

The estimates of 5-year risk for individuals in the test set range from 0.121 to 0.973.

Question 3

In this question we will look at two measures of calibration: mean calibration and the calibration slope. These are obtained for the test data. The code is given, but you can probably do all of this without looking at it.

Mean calibration

- Calculate the mean estimated 5-year risk in the test data, and compare this with the Kaplan-Meier estimate of risk up to 5 years.
- What is the ratio of the observed to the expected risk at 5 years?
- The solution shows how to obtain a 95% confidence interval for this ratio.

Calibration slope

- Calculate the linear predictor from the risk model for each individual.
- Perform a Cox regression of the outcome on this linear predictor.
- What is the resulting coefficient? This is the calibration slope.

SOLUTIONS

- Mean calibration: The mean 5-year risk estimated from the model is 0.440. The Kaplan-Meier estimate of 5-year risk is 0.450. The ratio of these is 1.022, with 95% CI 0.957 to 1.009. The ratio of observed to expected 5-year risk is close to 1, and the 95% CI includes 1. These results tell us that the mean calibration is good.
- Calibration slope: The coefficient for the linear predictor in a Cox regression on the linear predictor values LP is 1.10, with 95% CI 0.950 to 1.241. The estimated calibration slope is close to 1, and the 95% CI includes 1. This is another indicator that the calibration is good.

Question 4

Follow the steps below to create a calibration plot:

- Divide the predicted risks in the test data into 10 equal-sized groups (by placing cut-points at the deciles).
- Calculate the mean of the predicted risks for the people in each of the 10 groups.

- Obtain the Kaplan-Meier estimate of 5-year risk in each group.
- Plot the Kaplan-Meier estimates against the mean predicted risks.

Discuss what the plot shows.

SOLUTIONS

The observed risks within the 10 groups lie approximately on the 45-degree line when plotted against the mean predicted risks within the 10 groups. This indicates good calibration. The plot shows that people with low predicted risk were also observed to have low risk (looking at the Kaplan-Meier estimates of risk in the low predicted risk groups), and people with high predicted risk were also observed to have high risk (looking at the Kaplan-Meier estimates of risk in the high predicted risk groups).

Question 5

In this question we create a smooth version of the calibration plot in Question 4. Use the code provided to implement the following steps (all in the test data):

- Calculate $Z_i = \log\{-\log(1 - \widehat{R}_i(5))\}$ for each person in the test data, where $\widehat{R}_i(5)$ is the estimated 5-year risk.
- Perform a Cox regression including Z_i as the only covariate, using a spline (this is done using a restricted cubic spline in the code). This is called the ‘refitted model’.
- Obtain estimates of risk up to 5 years for each person based on the refitted model including Z_i .
- Plot the estimated risks from the refitted model against the estimated risks from the main prediction model.

SOLUTIONS

This plot is a smoother version of that seen in the previous question, and it shows that the calibration is good because the smoothed black curve lies close to the 45-degree line.

Question 6

In this question we look at measures of discrimination.

- Use the code provided to obtain estimates of Harrell’s c-index.
- Interpret the estimate.

There is no clear way of obtaining Uno's c-index from Stata

SOLUTIONS

The estimate of Harrell's c-index found in Stata is 0.698. In R the estimate of Harrell's c-index is 0.682 and the estimate of Uno's c-index is 0.681. The training and test sets are different in R and Stata as we created them randomly so we do not expect the estimates of Harrell's c-index from R and Stata to be identical. Uno's c-index (not available in Stata) accounts for censoring using weights, whereas Harrell's c-index ignores censoring. In this example the numbers are very similar. The estimated c-index is fairly low, indicating that the model does not have very good discrimination, in other words the ability of the model to rank people correctly in terms of their risk is not that good.

Question 7

Lastly we look at the Brier score, which is a measure of overall performance.

- Use the code provided to obtain an estimate of the Brier score and the scaled Brier score.
- Interpret the estimates. What does the scaled Brier score compare?

SOLUTIONS

The Brier score based on the Cox model is 0.204 and the Brier score based on the null model (with no predictors) is 0.247. The scaled Brier score ('IPA') is -21.3, indicating that the Brier score is reduced by 21.3% for the prediction model compared with the null model.

Question 8

So far we fitted the prediction model in the training data and evaluated the model performance in the test data. A better approach is to use k-fold cross validation. As a reminder, this involves the following steps, for 10-fold cross-validation:

- Divide the full data (i.e. before it was split in to training and test sets) randomly into approximately 10 equal-sized groups ('folds').
- Exclude fold 1 and fit the model in folds 2-10. Then obtain risk predictions in fold 1.
- Exclude fold 2 and fit the model in the remaining 9 folds. Then obtain predictions in fold 2.
- ...and so on, excluding each fold one by one. This results in a cross-validated risk prediction for each individual.

- Measures of predictive performance are then obtained using the cross-validated risk predictions on the full data.

See the code for an example of using 10-fold cross-validation.

Question 9

- Suppose you were happy to proceed with your model on the basis of the above evaluations of predictive performance. Obtain and write down a formula that could be used to obtain a prediction for a new individual.

SOLUTIONS

At the end, the chosen model should be refitted using all the data, i.e. using the training and test data combined. The formula for obtaining 5-year risk predictions is:

$$\widehat{R}(5) = 1 - \widehat{S}_0(5)^{\exp(LP)}$$

where the baseline survival probability at time 5 years is $\widehat{S}_0(5) = 0.08726972$ and the linear predictor LP is (with coefficients presented to 3 decimal places): $LP = -0.315 \times \text{hormon} - 0.330 \times \text{chemo} - 0.004 \times \text{age} + 0.047 \times \text{meno} + 0.361 \times I(\text{size} = 20 - 50) + 0.591 \times I(\text{size} > 50) + 0.349 \times I(\text{grade} = 3) - 1.916 \times \text{enodes} - 0.024 \times \text{pgr100} - 0.022 \times \text{er100} - 0.163 \times I(\text{year} > 1990)$.